

Foundations of Research

90 minutes · 5 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

What separates research from other ways of knowing, the main types of research, and the philosophical positions that quietly shape every methodological choice you will make later.

Learning objectives

- Define research and distinguish it from opinion, audit and routine reporting
- Classify research by purpose, approach and time dimension
- Explain the difference between positivist, interpretivist and pragmatic paradigms
- Trace how a paradigm constrains the designs available to you
- Describe the stages of the research process end to end

What research is, and is not

- Research is systematic, planned enquiry that produces evidence capable of being checked by someone else
- Three features distinguish it: a clear question, a documented method, and conclusions that follow from the data
- It is not: summarising what others said, confirming an opinion you already hold, or collecting data because it is available
- Audit asks "are we meeting an agreed standard?"; research asks "what is the case, and why?"
- The test of research is reproducibility — could a competent stranger follow your method and reach comparable conclusions?

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A great deal of published work fails this test. Reviewers increasingly ask for the protocol, the raw data and the analysis code precisely because the written method alone often is not enough to reproduce.

Classifying research

- By purpose: exploratory (what is going on here?), descriptive (what is the pattern?), explanatory (why does it happen?), evaluative (did it work?)
- By approach: quantitative (numbers, measurement, statistical inference), qualitative (meaning, context, interpretation), mixed methods (both, deliberately integrated)

- By time: cross-sectional (a snapshot), longitudinal (repeated over time), retrospective (looking back), prospective (following forward)
- By setting: laboratory, field, clinical, desk-based or secondary analysis
- By application: basic (extends knowledge) versus applied (solves a defined problem)

Paradigms: the assumptions underneath the method

- Ontology asks what reality is: a single objective reality, or multiple realities constructed by participants?
- Epistemology asks how we can know it: by measuring from the outside, or by interpreting from within?
- Positivism assumes an objective reality that can be measured — it leads naturally to experiments, surveys and statistical inference
- Interpretivism assumes reality is socially constructed — it leads to interviews, observation and thematic interpretation
- Pragmatism asks what works for the question at hand, and is the usual justification for mixed methods
- Critical and transformative paradigms foreground power and aim at change, common in participatory and action research

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You do not need to win the philosophical argument. You do need your paradigm, question, design and analysis to be consistent with one another — examiners notice when they are not.

The research process

- Identify a problem worth solving and confirm it is not already solved
- Review the literature and locate the gap precisely
- State the question, aims, objectives and (if quantitative) hypotheses
- Choose a design, population, sample and instruments
- Obtain ethical approval before collecting anything
- Collect, manage and analyse data
- Interpret against the literature, then report and disseminate
- The process is iterative in practice — the literature review continues until you submit

Worked examples

Audit or research? The same data, two different studies

A district hospital notices that 1 in 5 women who book antenatal care never return for a second visit. The medical director asks a midwife to "look into it".

1. If she compares the return rate against the national target of 4+ visits, that is audit — the standard already exists and she is measuring compliance.
2. If she asks why women do not return, and the answer is not already known, that is research.
3. The same appointment records serve both. What separates them is the question, not the data.
4. The research version needs ethical approval, a sampling strategy and a defined method; the audit does not.
5. She could do both: audit to establish the size of the problem, then research to explain it.

The point: Decide which one you are doing before you collect anything. Teams that start as audit and later try to publish are routinely told the work needed ethical approval it never obtained.

When the paradigm and the method do not match

A master's student writes that she takes an interpretivist position because "reality is socially constructed". Her method section then describes a 40-item Likert questionnaire administered to 300 nurses, analysed with chi-square tests.

1. Interpretivism treats meaning as constructed and context-specific, so it points towards interviews or observation.
2. A fixed-response questionnaire assumes the researcher already knows the categories that matter — a positivist assumption.
3. Neither choice is wrong on its own. The problem is that they contradict each other in one study.
4. Two honest fixes: keep the survey and state a post-positivist position, or keep interpretivism and switch to interviews.
5. A third option is a genuine mixed methods design, but then the integration has to be planned, not implied.

The point: Examiners read the paradigm statement and the method side by side. This mismatch is one of the most common reasons for a viva correction.

Key terms

Paradigm

A set of shared assumptions about the nature of reality and how it can be known, which shapes what counts as legitimate evidence.

Ontology

Assumptions about the nature of reality — whether it is single and objective, or multiple and constructed.

Epistemology

Assumptions about what knowledge is and how it can legitimately be acquired.

Basic research

Research undertaken to extend understanding, without a specific application in view.

Applied research

Research directed at solving an identified practical problem.

Reproducibility

The property that another researcher, following the reported method with the same data, obtains the same result.

Activity

In pairs, take a recent paper from your own field. Identify its purpose, approach, time dimension and implied paradigm. Then argue whether the design actually matches the paradigm the authors claim.

Further reading

- **Creswell & Creswell — Research Design (Ch. 1: The Selection of a Research Approach)** — The clearest short treatment of paradigm-to-design fit.
- **Bryman — Social Research Methods (Ch. 1-2)** — Strong on ontology and epistemology without becoming abstract.
- **Kothari — Research Methodology: Methods and Techniques** — Widely used across African and South Asian universities; good on classification.

Self-check questions

1. A hospital collects readmission rates each quarter and compares them against a national standard to check compliance. This is best described as:
 - a. Explanatory research
 - b. Clinical audit
 - c. Exploratory research

- d. A cohort study
2. A researcher believes patient experience of stigma cannot be captured by a scale because it differs between individuals and settings. Which paradigm is closest to this position?
- Positivism
 - Interpretivism
 - Post-positivism
 - Determinism
3. Which sequence correctly orders the research process?
- Collect data → review literature → state question → obtain ethical approval
 - State question → collect data → obtain ethical approval → analyse
 - Review literature → state question → obtain ethical approval → collect data
 - Obtain ethical approval → review literature → collect data → state question
4. A study follows a group of 400 newly diagnosed patients forward for three years, measuring outcomes annually. On the time dimension this is:
- Cross-sectional
 - Retrospective
 - Prospective longitudinal
 - Case-control
5. The main reason paradigm matters practically is that it:
- Determines which journals will accept the paper
 - Constrains which designs and forms of evidence are coherent for your question
 - Decides whether ethical approval is required
 - Sets the required sample size

Answers

- b.** Measuring performance against an agreed external standard is audit. It becomes research only when it asks a question whose answer is not already defined by the standard — for example, why readmission varies between wards.
- b.** Treating reality as constructed and specific to person and context is the interpretivist position, which leads towards interviews and observation rather than measurement instruments.
- c.** Ethical approval must precede data collection, and the question must be settled before approval can be sought. Collecting first and seeking approval afterwards is a common and serious error that can render data unusable.
- c.** Participants are enrolled and then followed forward with repeated measurement, which is prospective and longitudinal. A cross-sectional study would measure once.

5. **b.** A paradigm is not decoration. It governs what counts as evidence, and therefore which designs, instruments and analyses hang together. Mismatches between stated paradigm and actual method are a frequent examiner criticism.

Research Methods: Quantitative & Qualitative Design · ICBB-RM101 · Institute of Computational Biology and Bioinformatics (ICBB)



The Research Problem, Questions and Objectives

90 minutes · 5 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

How to move from a broad interest to a question that is specific enough to answer, and how aims, objectives, hypotheses and conceptual frameworks fit together.

Learning objectives

- Write a problem statement that establishes magnitude, consequence and gap
- Convert a broad topic into a focused, answerable research question
- Apply FINER and PICO to test whether a question is worth asking
- Distinguish aims, objectives, research questions and hypotheses
- Draw a conceptual framework linking your key variables or concepts

The problem statement

- A problem statement answers four things in order: what is the problem, how big is it, what follows if it is not addressed, and what is not yet known
- Magnitude needs evidence — prevalence, incidence, cost, or documented reports, each cited
- The gap is the pivot of the whole proposal: it must be a gap in knowledge, not merely a gap in your own reading
- A gap can be empirical (nobody has measured this), methodological (measured badly), theoretical (no adequate explanation) or contextual (established elsewhere but never tested here)
- Contextual gaps are legitimate and are often the strongest basis for research in Ghana and across Africa

IN PRACTICE

"No study has been done in my district" is only a gap if there is reason to expect the local answer might differ. Say why it might — different health system, different exposure, different population structure.

From topic to question

- Narrow along four axes: population, concept or exposure, outcome, and setting or time
- "Malaria in children" is a topic; "Does distance to a health facility predict delayed treatment for malaria in under-fives in the Upper East Region?" is a question
- A good question is answerable with the data and time you actually have

- If you cannot name the analysis that would answer it, the question is not yet specific enough
- Quantitative questions specify variables and their relationship; qualitative questions specify experience, process or meaning

Testing the question: FINER and PICO

- FINER — Feasible, Interesting, Novel, Ethical, Relevant
- Feasibility is the most commonly overlooked: participants, time, funding, skills and access
- PICO structures comparative quantitative questions: Population, Intervention or exposure, Comparison, Outcome
- Add T for timeframe (PICOT) when duration matters
- For qualitative work, SPIDER is a better fit: Sample, Phenomenon of Interest, Design, Evaluation, Research type

Framing the same topic three ways

Element	Quantitative (PICO)	Qualitative (SPIDER)
Who	Under-fives with confirmed malaria	Caregivers of under-fives treated for malaria
What	Distance greater than 5 km to facility	Experience of seeking and delaying care
Compared with	Distance of 5 km or less	Not applicable
Outcome	Treatment delayed beyond 24 hours	How caregivers account for delay
Analysis implied	Logistic regression, odds ratio	Thematic analysis of interviews

Aims, objectives and hypotheses

- The aim is one sentence stating the overall purpose — usually beginning "To determine...", "To explore...", "To evaluate..."
- Objectives break the aim into specific, ordered, measurable steps, each starting with an action verb
- Three to five objectives is usual; more than six normally signals an unfocused study
- A hypothesis is a testable statement of an expected relationship, stated for quantitative work only
- The null hypothesis states no difference or no association; the alternative states the expected effect
- Qualitative studies state research questions rather than hypotheses, because they do not set out to test a prediction

IN PRACTICE

Objectives must map one-to-one onto your analysis plan and, later, onto your results section. If an objective produces no result, or a result answers no objective, something has gone wrong.

Conceptual and theoretical frameworks

- A theoretical framework borrows an existing theory (Health Belief Model, Theory of Planned Behaviour, Social Cognitive Theory) to explain your phenomenon
- A conceptual framework is your own diagram of how the variables in your study relate
- It should distinguish independent, dependent, mediating, moderating and confounding variables
- Drawing it early exposes variables you would otherwise forget to measure
- The framework is also what tells you which variables must be adjusted for in analysis

Worked examples

Narrowing a topic until it can actually be answered

A student arrives with the topic "mental health of students in Ghana". It is a real problem, but as written it cannot be studied.

1. Population: which students? Settle on final-year nursing students at one university.
2. Exposure or concept: what about them? Choose academic workload, measured in timetabled contact hours.
3. Outcome: what will you measure? Depressive symptoms on the PHQ-9.
4. Setting and time: one campus, one semester.
5. The question becomes: "Is timetabled workload associated with PHQ-9 score among final-year nursing students at [University] in the 2026 second semester?"
6. Now name the analysis: linear regression of PHQ-9 score on contact hours, adjusted for age, sex and financial stress. If you cannot name it, the question is still too vague.

The point: The test of a finished question is that it dictates the analysis. If two people would analyse your data differently, the question is not specific yet.

An objective that produces no result

A proposal lists four objectives. The third is "to explore the perceptions of health workers". The study is a cross-sectional survey with a closed-response questionnaire.

1. "Explore perceptions" implies qualitative work — interviews or focus groups.
2. The instrument collects no free text, so nothing in the data can answer this objective.
3. At write-up the student will either drop the objective, or pad the results with a frequency table that does not address it.
4. Either outcome invites the examiner question: why did you state an objective you never met?
5. The fix at proposal stage is cheap: rewrite it as "to describe health workers' reported barriers", or add a qualitative strand and design for it.

The point: Map every objective to the specific analysis that will answer it before submitting the proposal. An unanswerable objective costs nothing to fix now and a chapter of trouble later.

Key terms

Research gap

A specific limitation in existing knowledge that the proposed study will address.

Aim

A single statement of the overall purpose of the study.

Objective

A specific, measurable step that contributes to achieving the aim.

Null hypothesis

The testable statement that no difference or association exists in the population.

Conceptual framework

A diagram or model of the presumed relationships among the study variables.

Confounder

A variable associated with both exposure and outcome that can produce a spurious association.

Activity

Write a one-paragraph problem statement and one research question for your own study. Exchange with a partner, who must identify the gap type and name the analysis your question implies. If they cannot, revise it.

Further reading

- **Hulley et al. — Designing Clinical Research (Ch. 2)** — Origin of the FINER criteria; concise and practical.
- **Cooke, Smith & Booth (2012) — "Beyond PICO: The SPIDER Tool"** — The qualitative counterpart to PICO.
- **Ravitch & Riggan — Reason & Rigor** — The best treatment of conceptual frameworks as an argument rather than a diagram.

Self-check questions

1. Which of the following is a research question rather than a topic?
 - a. Adolescent mental health in Ghana
 - b. The effect of social media on young people
 - c. Is daily social media use of more than three hours associated with depressive symptoms among senior high school students in Accra?
 - d. Investigating depression and technology
2. In FINER, the criterion most often underestimated by postgraduate students is:
 - a. Interesting
 - b. Novel
 - c. Feasible
 - d. Relevant
3. Which statement is a correctly framed null hypothesis?
 - a. Distance to a health facility increases treatment delay.
 - b. There is no association between distance to a health facility and treatment delay.
 - c. Does distance to a health facility affect treatment delay?
 - d. Treatment delay is an important problem in rural districts.
4. A study on a topic well established in Europe but never examined in West Africa is justified by which type of gap?
 - a. Methodological gap
 - b. Theoretical gap
 - c. Contextual gap
 - d. No gap exists
5. The relationship between objectives and the results section should be:

- a. Independent — results may cover whatever the data show
 - b. One-to-one — each objective yields a result and each result answers an objective
 - c. Objectives are for the proposal only and need not appear later
 - d. Results should exceed the objectives to show thoroughness
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Answers

1. **c.** Only option C specifies population, exposure, outcome and setting, and it implies a definite analysis. The others name areas of interest without stating what would be measured or compared.
 2. **c.** Feasibility — access to participants, time, funding and skills — is where most over-ambitious proposals fail. A question can be novel, ethical and relevant and still be impossible within a one-year programme.
 3. **b.** The null hypothesis asserts the absence of an association. Option A is the alternative hypothesis, C is a question, and D is a background claim.
 4. **c.** This is a contextual gap. It is legitimate, but the proposal must explain why the local answer could plausibly differ — different health system, exposure profile or population structure — rather than merely noting absence.
 5. **b.** Objectives are the skeleton of the study. An objective with no result suggests work not done; a result answering no objective suggests analysis that drifted, which invites accusations of fishing for findings.
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The Literature Review

120 minutes · 5 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

What a literature review is for, the main types and when each is appropriate, and how to build one that argues rather than merely lists.

Learning objectives

- Explain the four functions a literature review performs in a research project
- Distinguish narrative, scoping, systematic, rapid and integrative reviews
- Select the review type appropriate to your purpose and resources
- Organise a review thematically rather than source by source
- Use a synthesis matrix to move from reading to argument
- Recognise and avoid the most common review failures

What the review is for

- To establish what is already known, so you do not repeat settled work
- To locate and justify the gap your study will fill
- To inform your method — which designs, instruments and measures others found workable
- To supply the framework against which you will later interpret your own findings
- A review is an argument leading to your question, not a catalogue of everything published

IN PRACTICE

The single most common examiner criticism is a review that summarises twenty studies in twenty paragraphs and never says what they collectively mean. If a paragraph begins with an author name rather than an idea, suspect this failure.

Types of review

- Narrative review — broad, expert overview; flexible but vulnerable to selection bias
- Scoping review — maps the extent and nature of evidence and identifies gaps; ideal when a field is new or heterogeneous
- Systematic review — answers a focused question using a pre-registered protocol and exhaustive search; the highest standard, and the most work

- Meta-analysis — statistical pooling of results across studies; only valid when studies are sufficiently comparable
- Rapid review — systematic methods with deliberate, documented shortcuts, used under time pressure
- Integrative review — combines experimental and non-experimental evidence, common in nursing and health services research

Choosing a review type

If your purpose is...	Use	Typical effort
Introduce a thesis chapter	Narrative or integrative	4-8 weeks
Map an emerging or scattered field	Scoping review	3-6 months
Answer a focused effectiveness question	Systematic review	6-12 months
Pool effect estimates numerically	Systematic review with meta-analysis	9-18 months
Inform an urgent policy decision	Rapid review	4-12 weeks

Building the review

- Define the scope first: concepts, populations, date range, languages and study types, with a reason for each limit
- Search systematically even for a narrative review — record databases, terms and dates so the search can be repeated
- Screen in two passes: title and abstract, then full text against explicit criteria
- Extract into a synthesis matrix: rows are studies, columns are the themes, methods and findings you care about
- Reading the matrix down the columns rather than across the rows is what turns reading into synthesis
- Organise the written review by theme, debate or chronology of ideas — never by author

Appraising what you find

- Not all evidence carries equal weight; state how you judged quality
- Evidence hierarchies place systematic reviews and randomised trials above cohort, case-control and cross-sectional designs for effectiveness questions
- Hierarchies apply to questions of effectiveness, not to questions of meaning or experience — a qualitative study is not "low quality" evidence for how patients understand illness
- Use a structured tool: CASP checklists, the Newcastle-Ottawa Scale, or Cochrane Risk of Bias 2

- Watch for predatory journals — check indexing, editorial board, and whether the publisher appears in DOAJ or on Cabells lists
- Note conflicts of interest and funding sources

Common failures

- Listing rather than synthesising — the "annotated bibliography" review
- Relying only on what is freely available, producing a review biased towards open-access journals
- Citing abstracts without reading the full paper, and repeating a misreading from a secondary source
- Ignoring evidence that contradicts your position, which reviewers will notice
- Reviewing only recent work and losing the origin of a debate
- Letting the review stop when data collection begins — it must be updated before submission

Worked examples

Turning five papers into one paragraph

A student has read five studies on adherence to hypertension medication and has written five paragraphs, one per study, each beginning with an author name.

1. Build a matrix: rows are the five studies, columns are design, setting, sample, adherence measure and main finding.
2. Reading down the "adherence measure" column shows three used self-report and two used pill counts.
3. Reading down "main finding" shows the self-report studies all report higher adherence than the pill-count studies.
4. That comparison is the paragraph: "Reported adherence varies with how it was measured, with self-report consistently higher than objective measures (refs) — a pattern that complicates comparison across settings."
5. No sentence in that paragraph has an author as its subject. The studies are evidence for a claim, not the topic of the writing.

The point: Synthesis is what you see reading down the columns. Summary is what you get reading across the rows. Reviewers can tell instantly which one you did.

Choosing the review your timeline can survive

A student with nine months until submission proposes "a systematic review and meta-analysis of interventions to reduce malaria in under-fives across sub-Saharan Africa".

1. A proper systematic review needs a protocol, two independent screeners, dual data extraction and risk-of-bias assessment.
2. That question would return several thousand records to screen at title and abstract.
3. She is working alone, so the second screener does not exist — and without one it is not a systematic review by PRISMA standards.
4. A scoping review of the same area is defensible, achievable solo, and still publishable.
5. Alternatively, narrow the question hard: one intervention, one age band, one country — which can make a systematic review feasible.

The point: Pick the review type from your resources and question, not from which label sounds most impressive. A finished scoping review beats an abandoned systematic one.

Key terms

Synthesis matrix

A table with studies as rows and themes or attributes as columns, used to compare across sources rather than summarise each in turn.

Scoping review

A structured review that maps the extent, range and nature of evidence on a topic and identifies gaps.

Grey literature

Material outside commercial publishing — theses, government and NGO reports, working papers, conference abstracts.

Publication bias

The tendency for studies with positive or significant results to be published more often, distorting the apparent evidence.

Predatory journal

A journal charging publication fees while providing little or no genuine peer review or editorial service.

Saturation (in reviewing)

The point at which additional searching returns no new relevant concepts.

Activity

Take five papers you have already read. Build a synthesis matrix with columns for design, sample, setting, key finding and limitation. Then write one paragraph that compares across the column "key finding" without naming an author as the subject of any sentence.

Further reading

- **Booth, Sutton & Papaioannou — Systematic Approaches to a Successful Literature Review** — The best single book on choosing and executing a review type.
- **Arksey & O'Malley (2005) — scoping review framework** — The foundational scoping methodology, refined later by Levac and by JBI.
- **Page et al. (2021) — PRISMA 2020 statement** — The reporting standard; read it before you start searching, not after.

Self-check questions

1. A review that maps how much and what kind of research exists on a broad, emerging topic, without pooling results, is a:
 - a. Meta-analysis
 - b. Scoping review
 - c. Rapid review
 - d. Narrative review
2. The clearest sign that a literature review is listing rather than synthesising is that:
 - a. It cites more than fifty sources
 - b. Most paragraphs begin with an author name and describe one study each
 - c. It includes grey literature
 - d. It is organised chronologically
3. Judging a qualitative study of patient experience as "weak evidence" because it sits low on the evidence hierarchy is:
 - a. Correct — hierarchies apply to all research
 - b. A misapplication, because hierarchies rank evidence for effectiveness questions
 - c. Correct only if the sample is under 30
 - d. Correct because qualitative work is not generalisable
4. Excluding studies that are not freely available online risks introducing:

- a. Recall bias
 - b. Access bias, skewing the review towards open-access sources
 - c. Confounding
 - d. Attrition bias
5. When should the literature review be considered finished?
- a. When the proposal is approved
 - b. When data collection begins
 - c. Shortly before submission, after a final updating search
 - d. After fifty sources have been cited
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Answers

1. **b.** Scoping reviews are designed to map extent, range and nature of evidence and to identify gaps. They deliberately do not pool effects or usually appraise quality in the way a systematic review does.
 2. **b.** Paragraph-per-study structure signals summary rather than synthesis. A synthesising review organises paragraphs around ideas and cites several studies within each to support, qualify or contest a claim.
 3. **b.** Evidence hierarchies rank designs for answering questions about whether an intervention works. They say nothing about the quality of evidence for questions of meaning, experience or process, where qualitative designs are the appropriate choice.
 4. **b.** Restricting to what you can download for free is a selection problem. Use your institutional library, interlibrary loan, HINARI or AJOL, or write to the corresponding author, and state any unavoidable restriction explicitly.
 5. **c.** The field continues publishing while you work. A final updating search before submission protects you from the examiner who knows of a directly relevant paper published during your fieldwork.
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Literature Databases, Searching and Reference Management

150 minutes · 6 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

The databases that matter, how to build a search that is both sensitive and precise, how to document it for PRISMA, and how to manage references so the bibliography is never done by hand.

Learning objectives

- Select appropriate databases for a given topic and justify the choice
- Construct a Boolean search using controlled vocabulary and field tags
- Use MeSH terms, truncation, phrase searching and proximity operators
- Balance sensitivity against precision and explain the trade-off
- Document a search so that another researcher can reproduce it exactly
- Set up Zotero or Mendeley and generate a formatted bibliography automatically

The major databases

- PubMed / MEDLINE — biomedical and life sciences; free; indexed with MeSH controlled vocabulary
- Scopus and Web of Science — large multidisciplinary citation databases; subscription; best for citation tracking and bibliometrics
- Cochrane Library — controlled trials and systematic reviews; essential for effectiveness questions
- CINAHL — nursing and allied health; EMBASE — stronger European and pharmacological coverage than PubMed
- PsycINFO — psychology and behavioural sciences; ERIC — education
- African Journals Online (AJOL) — the largest collection of African-published scholarship, poorly covered by the big indexes
- Google Scholar — very broad and useful for finding grey literature and chasing citations, but the algorithm is opaque and results are not reproducible, so it should supplement rather than replace a structured search

IN PRACTICE

For work on African populations, searching only PubMed and Scopus will systematically miss regionally published evidence. AJOL, national journal portals and institutional repositories are not optional extras here.

Building the search

- Break the question into concepts — usually the P, I/E and O of PICO; do not search the whole question as a sentence
- For each concept, gather synonyms, spelling variants, abbreviations and older terminology
- Combine synonyms within a concept using OR; combine concepts using AND; use NOT sparingly, as it discards relevant records
- Truncation: malari* retrieves malaria, malarial, malariae
- Phrase searching: "mobile money" keeps the words adjacent
- Field tags restrict where a term is searched, e.g. [tiab] for title and abstract in PubMed
- Proximity operators (NEAR, W/n, ADJn) find terms near each other and vary by database

Controlled vocabulary

- MeSH is the National Library of Medicine's controlled vocabulary; each PubMed record is tagged by indexers
- Searching a MeSH term retrieves records regardless of the authors' wording
- Explosion automatically includes narrower terms beneath a heading in the tree
- Major topic restricts to records where the concept is a main focus
- MeSH indexing lags publication, so very recent papers are found only by free-text terms
- The reliable strategy combines both: MeSH OR free-text, for each concept
- Emtree in EMBASE and CINAHL Headings serve the same purpose in their databases

Sensitivity versus precision

Goal	Strategy	Cost
High sensitivity (systematic review)	Many synonyms, truncation, explode MeSH, no limits	Thousands of records to screen
High precision (quick scoping)	Few terms, major topic, title field, recent years	Relevant studies will be missed
Balanced (thesis chapter)	MeSH OR free-text per concept, sensible date limit	Manageable; state the limits used

Documenting the search

- Record for each database: its name, the platform, the date searched, the full strategy line by line, and the number of records returned
- The date matters because databases change daily and the search must be reproducible

- Report screening with a PRISMA flow diagram: identified, duplicates removed, screened, excluded with reasons, included
- Keep the exclusion reasons — reviewers ask for them
- Save the strategy in the database's own account feature and export it, rather than retyping later
- Set up email alerts so new hits arrive during your fieldwork

Reference management

- Zotero — free and open source, excellent browser capture, best default for most researchers
- Mendeley — free, good PDF annotation, owned by Elsevier
- EndNote — paid, common in institutions with a site licence, strong for very large libraries
- All three insert citations into Word or LibreOffice and reformat the entire bibliography when you change style
- Deduplicate on import, and fix metadata at import time — a wrong year propagates into every citation
- Use collections per chapter and tags per theme; attach the PDF and your notes to the record
- Back up the library, and sync it — a lost reference library near submission is a genuine disaster
- Never type a bibliography by hand: it wastes days and introduces errors that examiners do notice

IN PRACTICE

Set your citation style once at the beginning. Switching from APA to Vancouver later is a single dropdown in the reference manager, and a week of manual work without one.

Worked examples

Building a PubMed search line by line

The question is whether community health worker home visits reduce childhood malaria in rural Ghana. Typing the whole sentence into PubMed returns almost nothing useful.

1. Split into concepts: (1) community health workers, (2) home visits, (3) malaria in children.
2. Concept 1: "Community Health Workers"[Mesh] OR community health worker*[tiab] OR CHW[tiab] OR "village health volunteer*" [tiab]
3. Concept 2: "House Calls"[Mesh] OR home visit*[tiab] OR domiciliary[tiab]
4. Concept 3: "Malaria"[Mesh] AND ("Child"[Mesh] OR child*[tiab] OR under-five*[tiab])
5. Combine: #1 AND #2 AND #3. Record the date, the number of hits and the exact strings.
6. Then rerun the concepts in AJOL and Google Scholar — the Ghanaian and regional literature is thin in PubMed and you will otherwise miss it.

The point: Search concepts, never sentences. And a search you cannot reproduce next month, because you did not record it, is not a method — it is a memory.

What truncation actually costs you

A researcher searches nurs expecting to capture nurse, nurses and nursing. The result set balloons and is full of irrelevant records.*

1. nurs* also matches nursery, nurseries and nursed, none of which are wanted.
2. Truncating too early is the usual cause: nurs* cuts before the stem is distinctive.
3. nurse* OR nursing captures what is needed without the nursery records.
4. Check the effect: run both and compare the hit counts and a sample of titles.
5. Where a database offers it, a field tag narrows further — nurse*[tiab] rather than a free-text search of the whole record.

The point: Truncate at the point where the stem is still unambiguous, then look at what you actually retrieved. Wildcards are not free.

Key terms

MeSH

Medical Subject Headings — the controlled vocabulary used to index MEDLINE and PubMed records.

Boolean operators

AND, OR and NOT, used to combine search concepts.

Truncation

A wildcard symbol (usually *) retrieving all word endings from a common stem.

Sensitivity

The proportion of all relevant records that a search retrieves.

Precision

The proportion of retrieved records that are actually relevant.

PRISMA flow diagram

The standard figure reporting how records moved from identification through screening to inclusion.

Grey literature

Reports, theses and working papers published outside commercial channels and often outside major indexes.

Activity

Build a full PubMed strategy for your own question. Produce at least two concepts, each with MeSH and free-text synonyms combined by OR, then joined by AND. Record the date and the number of hits. Import the first twenty results into Zotero and generate a bibliography in APA, then switch it to Vancouver.

Further reading

- **PubMed User Guide — search field tags and MeSH** — Free, authoritative and short.
- **Bramer et al. (2018) — "A systematic approach to searching"** — A practical, reproducible method for building strategies.
- **Cochrane Handbook, Chapter 4 — Searching for studies** — The reference standard for exhaustive searching.
- **Zotero documentation — Quick Start Guide** — Enough to be productive in under an hour.

Self-check questions

1. To find records using either term for the same concept, you should combine "adolescent" and "teenager" with:
 - a. AND
 - b. OR
 - c. NOT
 - d. NEAR

2. The main advantage of searching with MeSH terms rather than free text alone is that:
 - a. MeSH searches are faster
 - b. Records are retrieved regardless of the wording the authors chose
 - c. MeSH always returns fewer results
 - d. MeSH includes the most recently published papers first
 3. A researcher studying maternal health services in Ghana searches only PubMed and Scopus. The most likely consequence is that they:
 - a. Retrieve too many irrelevant records
 - b. Miss regionally published African evidence indexed in AJOL and national repositories
 - c. Violate PRISMA reporting rules
 - d. Introduce recall bias
 4. A systematic review search should favour:
 - a. Precision, to keep screening manageable
 - b. Sensitivity, accepting a larger screening burden
 - c. Google Scholar alone, for breadth
 - d. Only papers published in the last five years
 5. Why must the date of each database search be recorded?
 - a. To calculate the response rate
 - b. Because databases are updated continuously, so the search is only reproducible as at a given date
 - c. It is required for ethical approval
 - d. To determine the citation style
 6. The strongest practical reason to use a reference manager is that it:
 - a. Improves the quality of your writing
 - b. Reformats every citation and the whole bibliography when the required style changes
 - c. Guarantees you avoid plagiarism
 - d. Finds relevant papers for you
-

Answers

1. **b.** OR broadens by retrieving records containing either synonym, so it combines terms within a concept. AND narrows and is used between different concepts.
2. **b.** Indexers assign MeSH consistently, so a MeSH search captures records however the authors phrased it. Its weakness is the opposite: very recent records are not yet indexed, which is why MeSH and free text are combined

with OR.

3. **b.** Coverage of African-published journals in the major international indexes is incomplete. AJOL, institutional repositories and national portals are needed to avoid systematically missing local evidence.
4. **b.** Systematic reviews aim to find all relevant studies, so they prioritise sensitivity and accept screening thousands of titles. Missing eligible studies biases the conclusion; screening extra records only costs time.
5. **b.** New records are added daily. Without the search date, another researcher rerunning your strategy cannot explain why their result count differs, and reproducibility is lost.
6. **b.** Style switching is the decisive practical gain — a dropdown instead of days of manual reformatting. It does not write for you, and it does not by itself prevent plagiarism.

Research Methods: Quantitative & Qualitative Design · ICBB-RM101 · Institute of Computational Biology and Bioinformatics (ICBB)



Quantitative Research Designs

120 minutes · 6 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

The main observational and experimental designs, what each can and cannot establish, and how to choose the weakest design that still answers your question honestly.

Learning objectives

- Describe cross-sectional, case-control, cohort and experimental designs
- Match a design to a research question and justify the choice
- Explain why association does not establish causation and what would
- Identify the principal biases threatening each design
- Explain randomisation, blinding and allocation concealment
- Recognise quasi-experimental designs and when they are the honest option

Observational designs

- Cross-sectional — exposure and outcome measured at the same moment; fast and cheap; gives prevalence but cannot establish which came first
- Case-control — starts from the outcome and looks backward at exposure; efficient for rare outcomes; vulnerable to recall and selection bias; yields an odds ratio
- Cohort — starts from exposure and follows forward to outcome; establishes temporal sequence; expensive and vulnerable to loss to follow-up; yields relative risk and incidence
- Ecological — units of analysis are groups rather than individuals; cheap but subject to the ecological fallacy
- Case series and case reports — descriptive only, no comparison group, hypothesis-generating at best

Choosing an observational design

Design	Starts from	Best for	Main weakness
Cross-sectional	A population at one time	Prevalence, associations	No temporal sequence
Case-control	The outcome	Rare outcomes, long latency	Recall and selection bias
Cohort	The exposure	Incidence, temporal sequence	Cost, attrition
Ecological	Groups	Population-level hypotheses	Ecological fallacy

Experimental designs

- The investigator assigns the exposure, which is what permits causal inference
- Randomisation distributes both known and unknown confounders across arms — its unique strength
- Allocation concealment stops the recruiter from knowing the next assignment, preventing selection bias at entry
- Blinding stops participants, providers or assessors from knowing the assignment, preventing performance and detection bias
- Randomised controlled trials may be parallel, crossover, factorial or cluster-randomised
- Cluster randomisation assigns groups (schools, clinics, villages) and requires a larger sample because responses within a cluster are correlated
- Analyse by intention to treat: participants are analysed in the arm to which they were assigned, whatever they actually received

IN PRACTICE

Per-protocol analysis, which counts only those who complied, quietly destroys randomisation and reintroduces confounding. Report it as a secondary analysis if at all.

Quasi-experimental designs

- Used when randomisation is impossible, unethical or politically unacceptable
- Non-equivalent control group — an untreated comparison group not formed by randomisation
- Interrupted time series — repeated measures before and after an intervention, strong when many time points exist
- Difference-in-differences — compares change over time between exposed and unexposed groups
- Stepped wedge — all clusters eventually receive the intervention, in a randomised order; ethically attractive for programme rollout
- These designs support weaker causal claims and must state the residual threats explicitly

Causation and bias

- Association is necessary but not sufficient for causation
- The Bradford Hill considerations — strength, consistency, temporality, biological gradient, plausibility and others — aid judgement but prove nothing individually
- Temporality is the only genuinely necessary condition: the cause must precede the effect
- Selection bias arises when those included differ systematically from those not included
- Information bias arises from measurement error — including recall bias and interviewer bias

- Confounding arises when a third variable explains the apparent relationship
- Confounding can be handled by design (randomisation, restriction, matching) or by analysis (stratification, regression adjustment)

Worked examples

One question, four designs, four different answers

Does using an insecticide-treated net reduce malaria in under-fives? The design chosen changes what can honestly be claimed.

1. Cross-sectional: survey households, record net use and current infection. Shows association only — sick children may have been given nets because they were sick.
2. Case-control: take children with malaria and matched controls, ask about past net use. Efficient, but caregivers of sick children recall exposure differently (recall bias).
3. Cohort: enrol net users and non-users, follow for a season. Establishes sequence, but families choosing nets differ from those who do not.
4. Cluster-randomised trial: randomise villages to receive nets. Balances known and unknown confounders — and answers the causal question.
5. The trial is strongest and also the slowest and most expensive; the cross-sectional study is fastest and answers least.

The point: Choose the weakest design that still answers your question honestly, then state plainly what it cannot establish. Overclaiming from a cross-sectional study is the most common criticism in peer review.

Why per-protocol analysis quietly breaks a trial

In a trial of a six-month TB treatment regimen, 15% of the intervention arm stopped taking the drug. The analyst proposes excluding them "so we measure the drug's true effect".

1. Ask who stopped. Typically those with worse side effects, more advanced disease, or less social support.
2. Those same characteristics predict worse outcomes independently of the drug.
3. Removing them leaves a healthier intervention arm compared against an intact control arm.
4. Randomisation guaranteed comparable groups at baseline; this exclusion destroys that guarantee.
5. Intention to treat keeps everyone in the arm they were assigned, and answers the question a clinician actually faces: what happens if I prescribe this?

The point: Intention to treat is the primary analysis. Report per-protocol as a secondary sensitivity analysis if at all, and never as the headline result.

Key terms

Incidence

The rate of new cases arising in a population over a defined period.

Prevalence

The proportion of a population having the condition at a point in time.

Odds ratio

The ratio of the odds of exposure in cases to the odds in controls; the standard measure from case-control studies.

Allocation concealment

Preventing those enrolling participants from knowing the upcoming assignment.

Intention to treat

Analysing participants in the arm to which they were randomised, regardless of adherence.

Ecological fallacy

Wrongly inferring something about individuals from a relationship observed between groups.

Confounding

Distortion of an exposure-outcome association by a third variable related to both.

Activity

You are asked whether a new community health worker programme reduced under-five mortality. Randomisation is refused by the district. Propose two quasi-experimental designs, state what each can and cannot establish, and name the biases you would have to acknowledge.

Further reading

- **Rothman — Epidemiology: An Introduction** — Short, rigorous and readable on design and bias.
- **Hulley et al. — Designing Clinical Research** — Practical chapters on each design with worked examples.
- **Schulz, Altman & Moher (2010) — CONSORT 2010 statement** — What a trial must report, and therefore what it must do.

Self-check questions

1. A researcher surveys 500 adults, measuring current salt intake and current blood pressure on the same day. The fundamental limitation is that:

- a. The sample is too small
 - b. Temporal sequence cannot be established, so it is unclear which came first
 - c. An odds ratio cannot be calculated
 - d. Ethical approval is not possible
2. For a rare cancer with a 20-year latency, the most efficient design is:
- a. Prospective cohort
 - b. Case-control
 - c. Randomised controlled trial
 - d. Cross-sectional survey
3. The unique advantage of randomisation over statistical adjustment is that it:
- a. Increases the sample size
 - b. Balances unknown and unmeasured confounders as well as known ones
 - c. Removes the need for a control group
 - d. Guarantees blinding
4. In a trial, participants who stopped taking the drug are excluded from analysis. This:
- a. Is correct practice, known as intention to treat
 - b. Breaks randomisation and can reintroduce confounding
 - c. Improves the precision of the estimate
 - d. Is required by CONSORT
5. Countries with higher average chocolate consumption have more Nobel laureates per capita. Concluding that chocolate improves individual cognition commits:
- a. Recall bias
 - b. The ecological fallacy
 - c. Selection bias
 - d. Attrition bias
6. Which condition is genuinely necessary for a causal claim?
- a. A large sample
 - b. Statistical significance
 - c. Temporality — the cause precedes the effect
 - d. Biological plausibility

Answers

1. **b.** This is a cross-sectional design. Exposure and outcome are measured simultaneously, so it can show association but cannot establish that the exposure preceded the outcome.
2. **b.** A cohort study would need an enormous sample followed for decades to accumulate enough cases. Case-control starts from existing cases and looks backward, making rare, slow-developing outcomes tractable.
3. **b.** Regression can only adjust for confounders you measured. Randomisation distributes confounders you never thought of, which no analysis can do.
4. **b.** Excluding non-adherers is per-protocol analysis. Adherence is often related to prognosis, so removing those participants destroys the comparability randomisation created. Intention to treat is the primary analysis.
5. **b.** The data describe countries, not people. Inferring an individual-level relationship from a group-level association is the ecological fallacy — and here national wealth confounds both variables.
6. **c.** Temporality is the one Bradford Hill consideration that cannot be dispensed with. The others strengthen a causal argument but none is individually necessary or sufficient.



Qualitative Research Designs

120 minutes · 6 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

The major qualitative traditions, the data collection methods attached to each, and how rigour is established when the aim is understanding rather than measurement.

Learning objectives

- Distinguish phenomenology, grounded theory, ethnography, case study and narrative inquiry
- Match a qualitative tradition to a research question
- Design an interview guide and a focus group schedule
- Explain sampling logic in qualitative work and the concept of information power
- Apply the criteria of trustworthiness and explain reflexivity
- Recognise when a qualitative design is the right answer and when it is an evasion

The major traditions

- Phenomenology — describes the lived experience of a phenomenon; asks what it is like to live through this; small purposive samples, in-depth interviews
- Grounded theory — builds theory from data through constant comparison and theoretical sampling; data collection and analysis proceed together
- Ethnography — describes culture and practice through prolonged immersion and participant observation; fieldnotes are primary data
- Case study — examines a bounded system in depth using multiple data sources; may be single or multiple case, and is strong for "how" and "why" questions
- Narrative inquiry — treats the story itself as the unit of analysis, attending to how people construct accounts of their lives
- Descriptive qualitative — a legitimate, atheoretical option when the aim is a straightforward account of views and experiences

IN PRACTICE

Choosing a tradition commits you to its sampling, data collection and analysis. Claiming grounded theory and then running six interviews with thematic analysis is a mismatch examiners look for specifically.

Data collection

- In-depth interviews — best for sensitive topics and individual experience; typically 45-90 minutes
- Focus groups — 6-10 participants; strong for surfacing shared norms and disagreement; poor for sensitive or stigmatised topics
- Participant and non-participant observation — captures what people do rather than what they say they do
- Document analysis — policies, records, media; unobtrusive and useful for institutional questions
- Photovoice and participatory methods — shift some control to participants and suit community-based work
- Triangulation combines sources or methods so that a finding does not rest on one type of evidence

Interview guides

- Structure the guide in funnels: open broadly, then probe specifics, then close
- Ask open questions — "tell me about...", "walk me through...", "what was that like?"
- Avoid leading questions, double-barrelled questions and jargon
- Plan probes in advance: "can you say more?", "what happened next?", "can you give an example?"
- Six to ten main questions is usually enough for a 60-minute interview
- Pilot the guide and expect to revise it — this is a strength, not a failure
- In focus groups, plan how you will manage a dominant speaker and draw out quiet participants

Sampling and sample size

- Sampling is purposive: participants are chosen because they can illuminate the question, not to represent a population statistically
- Maximum variation sampling seeks diversity; homogeneous sampling seeks depth within a defined group
- Theoretical sampling, specific to grounded theory, selects the next participant based on what the emerging analysis requires
- Snowball sampling reaches hidden populations but tends to stay within one network
- Saturation is the point where new data generate no new codes or themes
- Information power is the more defensible modern concept: narrow aim, dense sample specificity, strong theory and strong dialogue all reduce the number needed
- Typical ranges: 6-12 for a homogeneous phenomenological study, 20-30 for grounded theory, but justify rather than cite a number

IN PRACTICE

Declaring saturation after a fixed number of interviews planned in advance is not saturation. Saturation is something you observe during analysis and must be able to evidence.

Rigour and reflexivity

- Credibility — do the findings represent the participants' reality? Supported by member checking, triangulation and prolonged engagement
- Transferability — could findings apply elsewhere? Supported by thick description of context so readers can judge
- Dependability — would the process be consistent if repeated? Supported by an audit trail
- Confirmability — do findings follow from the data rather than the researcher's preferences? Supported by memos and reflexive notes
- Reflexivity means recording your own position — profession, assumptions, relationship to participants — and how it shaped the work
- Generalisability in the statistical sense is not the aim, and apologising for its absence misunderstands the design

Worked examples

The same topic in four qualitative traditions

The topic is the experience of women living with obstetric fistula. Four traditions would produce four genuinely different studies.

1. Phenomenology: what is it like to live with fistula? 8-12 in-depth interviews, focused on lived experience.
2. Grounded theory: how do women come to seek or avoid repair? Theoretical sampling, constant comparison, ending in a theory of care-seeking.
3. Ethnography: how does the community treat these women? Months of observation in a village, fieldnotes as primary data.
4. Case study: how did one repair programme work? Interviews, records, observation and policy documents on a single bounded programme.
5. Each demands its own sampling, data collection and analysis. You cannot claim one and execute another.

The point: Name your tradition and then live by its rules. "Grounded theory" followed by six interviews and a thematic analysis is the mismatch examiners look for first.

Rewriting an interview guide that will not work

A student's first interview guide opens with: "Don't you agree that the long waiting times and rude staff are the main problems at this clinic?"

1. It is leading — it supplies the answer the interviewer expects.
2. It is double-barrelled — waiting times and staff attitude are two different things.
3. It is closed — it can be answered "yes", ending the topic.
4. It opens on a complaint, before any rapport exists, which shapes everything that follows.
5. Rewrite: "Tell me about the last time you came to this clinic. Walk me through the day." Then probe: "What was the waiting like?", "How did you find the staff?"

The point: Open wide, then narrow with probes. Pilot the guide on two people — you will change it, and that is a sign the piloting worked, not that the guide was bad.

Key terms

Purposive sampling

Selecting participants deliberately for their capacity to inform the research question.

Saturation

The point at which further data collection yields no new codes, categories or themes.

Information power

The principle that the richer and more specific the sample and aim, the fewer participants are needed.

Reflexivity

Systematic attention to how the researcher shapes and is shaped by the research.

Thick description

Detailed account of context and meaning sufficient for a reader to judge transferability.

Triangulation

Using multiple sources, methods or investigators to strengthen confidence in a finding.

Audit trail

A documented record of decisions taken through the research, allowing the process to be followed.

Activity

Draft a six-question interview guide for your own topic. Swap with a partner, who must flag every leading, closed or double-barrelled question and rewrite one of them. Then role-play the first two questions and note where the guide fails in practice.

Further reading

- **Braun & Clarke — Successful Qualitative Research** — Practical, opinionated and reliable throughout.
- **Malterud, Siersma & Guassora (2016) — "Sample Size in Qualitative Interview Studies"** — The information power concept; short and worth reading in full.
- **Lincoln & Guba — Naturalistic Inquiry** — The origin of the trustworthiness criteria.
- **Creswell & Poth — Qualitative Inquiry and Research Design** — Compares the five traditions side by side.

Self-check questions

1. A researcher wants to build a theory of how nurses develop resilience, collecting and analysing data simultaneously and choosing later participants based on emerging categories. This is:
 - a. Phenomenology
 - b. Grounded theory
 - c. Ethnography
 - d. Narrative inquiry
2. Focus groups are a poor choice for which topic?
 - a. Community attitudes to a new clinic
 - b. Individual experiences of intimate partner violence
 - c. Shared norms about handwashing
 - d. Reactions to a proposed health policy
3. "Don't you think the long waiting times are the main problem at the clinic?" is flawed because it is:
 - a. Too open
 - b. Leading, and also closed
 - c. Double-barrelled only
 - d. Appropriately probing
4. A student writes: "Saturation was achieved as 15 interviews had been planned." This is unsatisfactory because:
 - a. Fifteen interviews is too few
 - b. Saturation is observed during analysis, not decided by a prior plan
 - c. Saturation applies only to grounded theory
 - d. Saturation requires at least 30 participants

5. Providing detailed description of setting, participants and context primarily supports which criterion?
- a. Credibility
 - b. Transferability
 - c. Dependability
 - d. Confirmability
6. A qualitative thesis states: "A limitation is that findings cannot be generalised to the wider population." The best response is that this:
- a. Is an appropriate limitation to acknowledge
 - b. Misapplies a quantitative criterion; transferability is the relevant concept
 - c. Means the sample size was too small
 - d. Requires a statistical power calculation
-

Answers

- 1. **b.** Simultaneous collection and analysis, constant comparison and theoretical sampling are the defining features of grounded theory, whose explicit aim is to generate theory.
 - 2. **b.** Stigmatised or sensitive individual experiences are unlikely to be disclosed in front of peers, and disclosure could put participants at risk. In-depth individual interviews are appropriate here.
 - 3. **b.** It supplies the answer and invites agreement, and it can be answered yes or no. "What is it like attending this clinic?" opens the topic without directing the response.
 - 4. **b.** Saturation is an empirical observation that new data are producing no new codes. Announcing it because a planned number was reached inverts the logic and cannot be evidenced.
 - 5. **b.** Thick description lets readers judge whether findings might apply in their own setting, which is how transferability works — the reader makes the judgement, not the researcher.
 - 6. **b.** Statistical generalisation was never the aim of the design. Framing its absence as a limitation suggests the researcher has not understood the logic of the method they used.
-

Mixed Methods and Sampling

120 minutes · 6 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

How quantitative and qualitative strands are combined deliberately rather than decoratively, and the full range of sampling strategies with the arithmetic of sample size.

Learning objectives

- Describe convergent, explanatory sequential and exploratory sequential designs
- Explain what integration means and why it is the defining feature of mixed methods
- Distinguish probability from non-probability sampling and state when each is appropriate
- Calculate a sample size for a prevalence study and for comparing two proportions
- Explain power, effect size and the consequences of an underpowered study
- Adjust a sample size for non-response and design effect

Mixed methods designs

- Convergent parallel — both strands run at the same time and are compared at interpretation; used when each answers a different facet of one question
- Explanatory sequential — quantitative first, then qualitative to explain surprising or unexplained results
- Exploratory sequential — qualitative first to develop an instrument or identify constructs, then quantitative to test at scale
- Embedded — one strand sits inside a larger design, such as a process evaluation within a trial
- Integration is the defining feature: strands must be brought together, not merely reported in adjacent chapters
- Integration techniques include joint displays, following a thread, and merging in a triangulation matrix

IN PRACTICE

A study that runs a survey and some interviews and reports them separately is not mixed methods; it is two studies in one document. The examiner's question is always: what did you learn from combining them that neither strand gave you alone?

Probability sampling

Institute of Computational Biology and Bioinformatics (ICBB) — Jesse Anak — icbb-gh.com

- Simple random — every unit has an equal chance; requires a complete sampling frame

- Systematic — every kth unit from a random start; simple in the field but fails if the list has a periodic pattern
- Stratified — the population is divided into strata and sampled within each; improves precision and guarantees representation of small subgroups
- Cluster — whole groups are sampled; cheaper when the population is geographically dispersed, but less precise
- Multistage — combines the above, and is what national surveys such as the DHS actually use
- Only probability sampling supports statistical generalisation to the population

Non-probability sampling

- Convenience — whoever is available; cheapest and weakest, with unknown bias
- Purposive — chosen for a defined characteristic; the standard for qualitative work
- Quota — fills preset category targets non-randomly; resembles stratification without randomisation
- Snowball — participants recruit others; necessary for hidden populations, but stays within networks
- Consecutive — every eligible patient over a defined period; the strongest non-probability option in clinical settings
- These cannot support statistical generalisation, and analyses reporting confidence intervals from them should be read cautiously

Sample size

- For a single proportion: $n = Z^2 p(1-p)/d^2$, where Z is 1.96 for 95% confidence, p the expected proportion, d the desired margin of error
- Worked example: $p = 0.30$, $d = 0.05$ gives $n = (1.96^2 \times 0.30 \times 0.70)/0.05^2 = 323$
- Using $p = 0.50$ when the proportion is unknown gives the largest, safest sample
- For comparing two proportions or means, size depends on the effect you wish to detect, the significance level and the power
- Power is the probability of detecting a real effect; 80% is conventional and 90% is preferable
- An underpowered study is not merely inconclusive — it is arguably unethical, since participants were exposed to burden for a result that could not have been informative
- Apply a finite population correction when the sample is a large fraction of a small population

Adjustment	When it applies	Formula
Non-response	Always, in surveys	$n_{\text{final}} = n / (1 - \text{expected non-response})$
Design effect	Cluster sampling	$n_{\text{final}} = n \times \text{DEFF (often 1.5-2.0)}$
Finite population	Sample exceeds ~5% of population	$n_{\text{adj}} = n / (1 + (n-1)/N)$
Attrition	Longitudinal designs	Inflate by expected loss to follow-up

Putting it together

- State the sampling frame explicitly, and say what it excludes
- Report the calculation with every input value, not just the final number
- Say who was excluded and why, and how non-response was handled
- Mixed methods studies size each strand by its own logic — power calculation for the quantitative strand, information power for the qualitative
- A nested qualitative sample is often drawn purposively from the quantitative sample, which strengthens integration

Worked examples

Calculating a sample size, then adjusting it honestly

A prevalence survey of anaemia among pregnant women in one district. Previous work suggests roughly 35% prevalence. You want a 5% margin of error at 95% confidence, sampling by health facility.

1. Base formula: $n = Z^2 p(1-p) / d^2 = (1.96^2 \times 0.35 \times 0.65) / 0.05^2$
2. That is $(3.8416 \times 0.2275) / 0.0025 = 0.8740 / 0.0025 = 350$ (rounding up).
3. Cluster sampling by facility: multiply by a design effect of 1.5 $\rightarrow 350 \times 1.5 = 525$.
4. Expect 10% non-response: $525 / 0.90 = 584$.
5. Final target: 584 women. In R: `ceiling(1.96^2 * 0.35 * 0.65 / 0.05^2 * 1.5 / 0.9)`
6. Report every input — Z, p, d, design effect and non-response — not just the number 584.

The point: Reviewers check the arithmetic. A sample size stated without its inputs reads as though it was chosen first and justified afterwards.

Two studies in one document is not mixed methods

A thesis reports a 600-household survey in Chapter 4 and 20 interviews in Chapter 5. The conclusion summarises both. The student calls it a mixed methods study.

1. Ask the integration question: what did combining them reveal that neither gave alone?
2. If the answer is "nothing — they are both about the same topic", it is two studies bound together.
3. Real integration would be: the survey found low clinic attendance in one sub-district; the interviews were sampled from there specifically to explain why.
4. Or a joint display: a table with survey findings in one column and the interview themes that explain or contradict them alongside.
5. The interview sample can be drawn from the survey respondents, which ties the strands together by design rather than by assertion.

The point: Integration is the defining feature. Plan the point where the strands meet at proposal stage — it cannot be added convincingly at write-up.

Key terms

Sampling frame

The list or procedure from which the sample is actually drawn.

Design effect (DEFF)

The factor by which variance increases because of cluster sampling rather than simple random sampling.

Power

The probability that a study will detect an effect of a specified size if it truly exists.

Effect size

The magnitude of a difference or association, independent of sample size.

Integration

The deliberate bringing together of quantitative and qualitative strands to produce insight neither gives alone.

Joint display

A table or figure presenting quantitative and qualitative results side by side to show how they relate.

Activity

Calculate the sample size for a prevalence study in your own field: choose p , set $d = 0.05$, then inflate for 15% non-response and a design effect of 1.5. Show every step. Then state, in one sentence each, the sampling frame and what it excludes.

Further reading

- **Creswell & Plano Clark — Designing and Conducting Mixed Methods Research** — The standard text; strong on integration and joint displays.
- **Fetters, Curry & Creswell (2013) — "Achieving Integration in Mixed Methods Designs"** — The clearest short account of what integration actually requires.
- **Lwanga & Lemeshow — Sample Size Determination in Health Studies (WHO)** — Free from WHO; tables for most common situations.

Self-check questions

1. A team surveys 600 households, finds unexpectedly low clinic attendance, then interviews 20 non-attenders to understand why. This design is:
 - a. Convergent parallel
 - b. Explanatory sequential
 - c. Exploratory sequential
 - d. Embedded
2. What most clearly distinguishes genuine mixed methods from simply doing two studies?
 - a. Using a larger sample
 - b. Deliberate integration of the strands to produce insight neither gives alone
 - c. Publishing both in the same journal
 - d. Collecting both types of data at the same time
3. For a prevalence study with expected proportion 0.30 and margin of error 0.05 at 95% confidence, the required sample is approximately:
 - a. 196
 - b. 323
 - c. 384
 - d. 512
4. Which strategy guarantees that a small subgroup is adequately represented?

- a. Simple random sampling
 - b. Stratified sampling
 - c. Convenience sampling
 - d. Systematic sampling
5. Cluster sampling requires a larger sample than simple random sampling because:
- a. Clusters are harder to find
 - b. Responses within a cluster are correlated, which increases variance
 - c. It is a non-probability method
 - d. Non-response is always higher
6. The strongest argument that a seriously underpowered study is unethical is that:
- a. It will not be published
 - b. Participants bore burden and risk for a study incapable of producing an informative answer
 - c. It wastes the researcher's time
 - d. It cannot obtain ethical approval
-

Answers

1. **b.** The quantitative strand comes first and the qualitative strand is used to explain its findings. That ordering and purpose define an explanatory sequential design.
 2. **b.** Integration is the defining criterion. Reporting a survey and some interviews in adjacent chapters, with no point where they speak to each other, is not a mixed methods design.
 3. **b.** $n = Z^2 p(1-p)/d^2 = (3.8416 \times 0.30 \times 0.70)/0.0025 = 0.8067/0.0025 \approx 323$. The familiar 384 comes from using $p = 0.50$, the most conservative value.
 4. **b.** Stratification divides the population first and samples within each stratum, so small subgroups are represented by design. Simple random sampling may by chance include very few of them.
 5. **b.** People within the same village, school or clinic tend to resemble one another, so each observation carries less independent information. The design effect quantifies this and inflates the required sample.
 6. **b.** Ethical justification for research rests on the prospect of useful knowledge. If the design could not have detected a meaningful effect, participants were exposed to burden with no realistic prospect of benefit.
-

Data Collection, Instruments and Measurement

120 minutes · 6 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

Turning abstract constructs into things you can actually record, and the validity and reliability evidence needed before anyone believes the numbers your instrument produces.

Learning objectives

- Operationalise a construct into measurable variables and indicators
- Identify levels of measurement and their analytical consequences
- Design questionnaire items that avoid the common wording faults
- Distinguish the main types of validity and reliability and how each is evidenced
- Plan and report a pilot study
- Choose between paper and electronic data capture and set up basic data quality controls

From construct to variable

- A construct is an abstraction — stigma, food insecurity, quality of care — that cannot be observed directly
- Operationalisation defines the observable indicators that will stand for it
- A conceptual definition says what you mean; an operational definition says exactly how it will be measured
- Prefer an existing validated instrument over inventing your own — validation takes a study of its own
- If you adapt an instrument, report what you changed and re-examine its reliability in your population
- Instruments validated elsewhere may behave differently after translation or in a different culture, and should be re-tested

IN PRACTICE

Translation requires forward translation, back translation by an independent translator, reconciliation by a panel, and cognitive interviewing with a few target participants. Translating a questionnaire yourself the night before is the single most common source of unusable data.

Levels of measurement

- Nominal — unordered categories (blood group, district); only counts and proportions are meaningful
- Ordinal — ordered categories with unequal or unknown intervals (Likert responses, disease stage)
- Interval — equal intervals but no true zero (temperature in Celsius)
- Ratio — equal intervals with a true zero (weight, income, count of visits)
- The level determines which summary statistics and which tests are legitimate
- Record data at the finest level available: exact age can always be collapsed into bands later, but bands can never be recovered into exact ages

Writing good items

- One idea per item — "Was the staff friendly and efficient?" is double-barrelled and unanswerable if one was and one was not
- Avoid leading wording, loaded terms and negatives, especially double negatives
- Match vocabulary to the least literate respondent you expect
- Make response options exhaustive and mutually exclusive; check for gaps and overlaps in age bands
- Offer "don't know" and "prefer not to answer" where forcing a choice would fabricate data
- Decide deliberately whether a Likert scale has a midpoint — removing it forces a direction, which may or may not be what you want
- Order matters: put sensitive items late, once rapport exists, and beware of earlier items priming later ones

Validity and reliability

- Face validity — does it look appropriate? The weakest form, but worth checking
- Content validity — does it cover the whole construct? Established by expert panel, often with a content validity index
- Construct validity — does it behave as theory predicts? Evidenced by convergent and discriminant correlations and factor analysis
- Criterion validity — does it agree with a gold standard, concurrently or predictively?
- Internal consistency — do items measure the same thing? Cronbach's alpha, where roughly 0.70-0.95 is acceptable and above 0.95 suggests redundant items
- Test-retest reliability — does it give the same answer on a stable characteristic over a short interval?
- Inter-rater reliability — do different observers agree? Reported as Cohen's kappa or an intraclass correlation
- An instrument can be reliable but not valid — consistently measuring the wrong thing

Property	Statistic	Rough benchmark
Internal consistency	Cronbach's alpha	0.70 - 0.95
Test-retest	ICC or Pearson r	≥ 0.70
Inter-rater agreement	Cohen's kappa	≥ 0.60 substantial
Content validity	CVI	≥ 0.78 per item
Construct validity	Factor loadings	≥ 0.40

Piloting and data capture

- Pilot on 10% of the intended sample, or at least 20-30 respondents, drawn from the same population but excluded from the main study
- The pilot tests comprehension, timing, flow, skip logic, coding and the analysis plan — run your intended analysis on pilot data
- Electronic capture (KoBoToolbox, REDCap, ODK, CommCare) enforces ranges and skip logic at the point of entry and removes a transcription step
- Paper remains appropriate where power, connectivity or literacy make devices impractical; plan double entry if so
- Build validation rules: allowed ranges, required fields, logical consistency checks between related items
- Train and standardise data collectors, then check inter-rater agreement rather than assuming it
- Anonymise at collection where possible, keeping any identifier link in a separate secured file

Worked examples

A questionnaire item pulled apart

A draft patient satisfaction survey includes: "How satisfied were you with the cost and quality of care? (Very satisfied / Satisfied / Dissatisfied)"

1. Double-barrelled: a patient happy with quality but not cost cannot answer truthfully.
2. The response options are unbalanced — two positive, one negative — which pushes answers upward.
3. There is no neutral option and no "not applicable", so someone who paid nothing must still choose.
4. Split into two items, each with a balanced 5-point scale from Very dissatisfied to Very satisfied.
5. Add "Not applicable" to the cost item for patients covered by the NHIS.
6. Pilot on 20 patients and ask them to say aloud what they thought each item meant.

The point: Every faulty item produces data you cannot interpret, and no analysis later can repair it. An hour spent on wording saves the whole variable.

What a translated instrument really requires

A validated English depression scale is to be used with Twi-speaking participants. The student translates it herself the week before fieldwork.

1. Forward translation by two independent bilingual translators, working separately.
2. Reconciliation of the two versions by a panel, resolving each difference deliberately.
3. Back translation into English by a third translator who has never seen the original.
4. Compare back translation with the original; where meaning has drifted, revise and repeat.
5. Cognitive interviewing with 5-10 target participants: ask what each item means to them in their own words.
6. Re-check reliability in the new population — the original Cronbach's alpha does not transfer.

The point: A self-translated instrument has unknown validity, which means the study built on it does too. This is one of the most common fatal flaws in multilingual research.

Key terms

Operationalisation

Specifying the observable indicators and procedures by which an abstract construct will be measured.

Construct validity

Evidence that an instrument measures the theoretical concept it claims to measure.

Cronbach's alpha

A coefficient of internal consistency among the items of a scale.

Cohen's kappa

A measure of agreement between raters that corrects for agreement expected by chance.

Skip logic

Rules routing a respondent past items that do not apply to them.

Double entry

Entering paper data twice independently and reconciling differences to catch transcription error.

Activity

Take one construct from your study and write three candidate items for it. Exchange with a partner, who must identify any double-barrelled, leading or ambiguous wording and check the response options are exhaustive and mutually exclusive. Then state which reliability evidence you would report and why.

Further reading

- **DeVellis — Scale Development: Theory and Applications** — The standard reference for constructing and validating scales.
- **Boateng et al. (2018) — "Best Practices for Developing and Validating Scales"** — Open access, health-focused, and unusually practical.
- **Beaton et al. (2000) — Guidelines for cross-cultural adaptation** — The forward/back translation process, done properly.

Self-check questions

1. "How satisfied were you with the cost and quality of care?" is faulty because it is:
 - a. Leading
 - b. Double-barrelled
 - c. Too open
 - d. Correctly written
2. A scale consistently gives the same score on repeat administration but correlates poorly with every established measure of the construct. It is:
 - a. Valid but unreliable

- b. Reliable but not valid
 - c. Both reliable and valid
 - d. Neither
3. Recording age in five-year bands rather than exact years is generally inadvisable because:
- a. Bands are harder for respondents to answer
 - b. Exact values can always be collapsed into bands later, but bands cannot be recovered
 - c. Bands are not a valid level of measurement
 - d. It always reduces the response rate
4. A Cronbach's alpha of 0.98 on a 20-item scale most likely indicates:
- a. Excellent construct validity
 - b. Redundant items that are near-duplicates of one another
 - c. Poor internal consistency
 - d. A sample that is too small
5. The primary purpose of a pilot study is to:
- a. Increase the total sample size
 - b. Test comprehension, timing, flow, coding and the analysis plan before full data collection
 - c. Publish preliminary findings
 - d. Satisfy the ethics committee
6. When adapting a validated English instrument for use in Twi, the minimum acceptable process includes:
- a. Direct translation by the researcher
 - b. Forward translation, independent back translation, expert reconciliation and cognitive interviewing
 - c. Using an online translation tool and checking it reads well
 - d. No adaptation is needed if respondents understand some English

Answers

1. **b.** It asks about two distinct things at once. A respondent satisfied with quality but not cost cannot answer, and their response cannot be interpreted. Split it into two items.
2. **b.** Consistency is reliability. Failing to relate to the construct it claims to measure is a validity failure. An instrument can measure the wrong thing very consistently.
3. **b.** Collecting at the finest available level preserves analytical options. Categorising at collection discards information permanently and forecloses analyses you may later need.

4. **b.** Very high alpha usually means items are asking essentially the same question in different words. It inflates apparent consistency while narrowing the construct actually covered.
5. **b.** A pilot is a rehearsal of the whole process, including running the intended analysis on pilot data. Pilot participants are excluded from the main study and do not add to its sample.
6. **b.** Cross-cultural adaptation is a defined procedure. Skipping it means the translated instrument's validity is unknown, and any psychometric evidence from the original version no longer applies.



Research Ethics and Integrity

120 minutes · 6 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

The principles governing research with human participants, how ethical review actually works, and the integrity questions — authorship, plagiarism, data fabrication and the use of AI — that decide whether your work stands.

Learning objectives

- State the four principles of research ethics and apply them to a scenario
- Explain what makes consent genuinely informed, voluntary and documented
- Assemble an ethics application and anticipate what a committee will query
- Distinguish anonymity from confidentiality and implement both
- Identify vulnerable groups and the additional safeguards they require
- Apply authorship criteria and recognise research misconduct
- State a defensible position on the use of AI tools in research

Principles and their origins

- The Nuremberg Code followed the Nazi medical experiments; the Declaration of Helsinki and the Belmont Report followed later abuses, including Tuskegee
- Autonomy — people decide for themselves whether to take part, having understood what is involved
- Beneficence — maximise benefit; Non-maleficence — minimise harm; the two are weighed together
- Justice — burdens and benefits are distributed fairly, and no group is used as a convenient subject pool for research that will benefit others
- Justice has particular weight in African research settings, where communities have historically borne research burden without sharing in benefit
- Ethical conduct is not a form to be completed; it is a continuing obligation through the study

Informed consent

- Information — purpose, procedures, duration, risks, benefits, alternatives, confidentiality limits, and who to contact
- Comprehension — in the participant's language, at an appropriate reading level, with opportunity to ask questions
- Voluntariness — free from coercion or undue inducement; compensation should cover cost and time, not act as an incentive that overrides judgement

- Documentation — signature, thumbprint or witnessed verbal consent, as appropriate to context and literacy
- The right to withdraw at any time, without giving a reason and without penalty, must be stated and honoured
- Children give assent while a parent or guardian gives consent; the age of assent varies by jurisdiction
- Consent is a process, not a signature: it should be revisited in longitudinal and ethnographic work
- Waiver of consent may be granted for anonymised secondary data or minimal-risk record review, but only by the committee

IN PRACTICE

Where a clinician recruits their own patients, or a lecturer their own students, the power relationship can make refusal feel impossible. Address this explicitly in the application — usually by having someone independent take consent.

Review and approval

- Institutional Review Boards and Research Ethics Committees review before any data are collected
- In Ghana, approvals commonly involve institutional RECs and, depending on the work, the Ghana Health Service Ethics Review Committee or the FDA
- A typical application requires the protocol, instruments, information sheet, consent form, translations, CVs and a data management plan
- Committees most often query: unclear risk-benefit, weak consent for vulnerable groups, inadequate data protection, undisclosed conflicts, and underpowered designs
- Approval is time-limited; amendments and extensions must be submitted, and adverse events reported
- Publishing without approval is a serious breach — many journals require the approval number, and retraction follows if it was never obtained

Confidentiality and data protection

- Anonymity means even the researcher cannot link data to a person; confidentiality means the researcher can but will not disclose
- Most studies offer confidentiality, not anonymity — say which, accurately
- De-identify at the earliest point; keep any linking key in a separate, encrypted, access-controlled file
- Ghana's Data Protection Act 2012 (Act 843) governs personal data, requiring lawful basis, purpose limitation and security
- State retention periods and how data will be destroyed or archived

- Qualitative data are especially re-identifiable — a role plus a district can identify someone even without a name
- Confidentiality has limits: disclosure of imminent harm or of certain crimes may compel action, and participants must be told this in advance

Integrity, authorship and AI

- Fabrication (inventing data), falsification (altering data) and plagiarism are the three classical forms of misconduct
- Self-plagiarism and duplicate publication are also breaches, even though the text is your own
- ICMJE authorship requires all four: substantial contribution, drafting or critical revision, final approval, and accountability for the work
- Supervisors are not automatically authors; funders and heads of department are not authors by virtue of position — gift authorship is a breach
- Agree the authorship order in writing at the start, not after the results arrive
- Declare funding and conflicts of interest; register trials prospectively and consider preregistering observational protocols
- Report negative and null results — suppressing them distorts the literature and wastes participants' contribution
- AI tools may assist with language, structure and code, but the researcher remains accountable for every claim; AI cannot be an author, must not be used to generate data or references, and its use should be disclosed as journals and institutions increasingly require
- Never paste identifiable participant data into a public AI service — that is a data protection breach regardless of the research value

Worked examples

Consent when the researcher holds power over the participant

A lecturer wants to survey her own final-year students about their experience of the programme. She will administer it herself, at the end of a lecture she teaches.

1. Students may reasonably fear that declining, or answering critically, could affect their marks.
2. That fear compromises voluntariness even if the lecturer would never act on it — the perception is enough.
3. Fix one: an independent colleague, with no assessment role, administers and holds the data.
4. Fix two: collect responses anonymously, and only after final marks are released.
5. The information sheet must state plainly that participation has no bearing on assessment.
6. Name the power relationship explicitly in the ethics application. Committees notice when it is left out.

The point: Whenever the researcher can affect the participant's grades, treatment or employment, address it in the design. Saying "participation was voluntary" does not make it so.

How a quotation identifies someone with no name attached

A report on medical errors quotes: "As the only male theatre nurse at the district hospital, I felt I could not challenge the surgeon."

1. No name appears, yet anyone in that district can identify the speaker immediately.
2. Role, gender, setting and seniority combine into an identifier even when each alone would not.
3. The consequences here are real — this nurse could face professional retaliation.
4. Options: remove the role detail, aggregate the setting to "a district hospital in the region", or paraphrase rather than quote.
5. Where the detail is essential to the point, go back to the participant and ask before publishing.
6. Say in the information sheet how quotations will be handled, so consent covers what you actually do.

The point: Anonymisation means removing what identifies, not just removing names. Read every quotation as someone in that workplace would read it.

Key terms

Autonomy

The principle that competent people decide for themselves whether to participate.

Beneficence

The obligation to maximise benefit and minimise harm to participants.

Justice

Fair distribution of the burdens and benefits of research across groups.

Assent

A child's own agreement to take part, alongside a guardian's consent.

Anonymity

Data cannot be linked to an individual by anyone, including the researcher.

Confidentiality

Identifiable data exist but access is restricted and disclosure prevented.

Gift authorship

Naming as author someone who has not met the criteria for authorship.

Equipoise

Genuine uncertainty about which trial arm is superior, the ethical basis for randomising.

Activity

You plan to interview nurses about errors they have witnessed. Draft the confidentiality paragraph of your information sheet, including its limits. Then list every way a reader might identify a participant from a quotation, and state how you would prevent each.

Further reading

- **World Medical Association — Declaration of Helsinki** — Short, foundational and worth reading in full at least once.
- **The Belmont Report (1979)** — The origin of respect for persons, beneficence and justice.
- **ICMJE Recommendations — authorship criteria** — The four-part test, and what disqualifies a contributor.
- **Ghana Data Protection Act 2012 (Act 843)** — The legal framework for personal data in Ghana.

Self-check questions

1. A researcher offers unemployed participants an amount equal to two weeks' wages to join a study with real risk. The principal concern is:
 - a. The budget is too large
 - b. Undue inducement compromising voluntariness

- c. A breach of confidentiality
 - d. Nothing — compensation is always acceptable
2. A study records names and phone numbers so participants can be followed up, storing them securely and never disclosing them. This offers:
- a. Anonymity
 - b. Confidentiality
 - c. Both
 - d. Neither
3. A head of department who provided no intellectual input asks to be listed as an author. Under ICMJE criteria this is:
- a. Acceptable, as departmental support is a contribution
 - b. Gift authorship, which is a breach of publication ethics
 - c. Acceptable if they are listed last
 - d. Required by most institutions
4. Data collection began before ethical approval was granted, though approval was later obtained. The likely consequence is:
- a. None, since approval was eventually granted
 - b. The data collected beforehand are unusable and publication may be refused or retracted
 - c. A brief delay only
 - d. The committee will backdate the approval
5. Which use of an AI tool is clearly unacceptable?
- a. Improving the grammar of a paragraph you wrote
 - b. Generating plausible-looking references you do not verify
 - c. Explaining a statistical concept to you
 - d. Helping debug your analysis code
6. A trial finds the new intervention no better than standard care, and the team decides not to publish. This is problematic because:
- a. Null results are always publishable in high-impact journals
 - b. Suppressing null findings distorts the evidence base and wastes participants' contribution
 - c. The funder requires a positive result
 - d. It has no ethical implications

Answers

1. **b.** Payment large enough to override a person's judgement about risk undermines voluntariness. Compensation should reasonably cover time and costs, not act as an incentive that makes refusal unrealistic.
2. **b.** Because identifiers exist and the researcher can link data to individuals, this is confidentiality. Anonymity would require that no one, including the researcher, could make that link.
3. **b.** Authorship requires substantial contribution, drafting or critical revision, final approval and accountability. Position or provision of facilities does not qualify; acknowledgement is the appropriate place.
4. **b.** Approval cannot be retrospective. Data gathered before it lack ethical authorisation, journals require the approval date, and discovering the discrepancy after publication commonly leads to retraction.
5. **b.** Generated references are frequently fabricated. Citing sources you have not verified and read is a fabrication of the evidentiary record, regardless of how the citation was produced.
6. **b.** Selective publication of positive findings produces publication bias, misleading later reviews and clinical decisions. Participants accepted burden on the understanding the knowledge would be shared.



Writing the Proposal, Rigour and Dissemination

120 minutes · 6 self-check questions

Author: **Jesse Anak** · Institute of Computational Biology and Bioinformatics (ICBB)

Assembling everything into a proposal that will be approved and funded, building in rigour rather than claiming it, and planning how findings will reach the people who can act on them.

Learning objectives

- Structure a research proposal to the standard expected by committees and funders
- Write a methods section detailed enough to be reproduced
- Build a realistic timeline and budget
- State limitations honestly without undermining the study
- Plan a data management strategy
- Design a dissemination plan that reaches participants and decision makers, not only journals

Proposal structure

- Title — specific, informative, usually naming design, population and setting
- Abstract — written last, covering background, aim, methods, expected outcomes; respect the word limit exactly
- Introduction and background — the problem, its magnitude, and the gap
- Literature review — the argument leading to your question
- Aim, objectives, research questions and hypotheses
- Methodology — design, setting, population, sampling, sample size, instruments, procedures, analysis plan
- Ethical considerations — approvals sought, consent, confidentiality, risk mitigation
- Timeline, budget, references, appendices with instruments and consent forms
- Reviewers commonly read the aim, then jump to sampling, sample size and analysis — those must survive isolated scrutiny

The methods section

- The test is reproducibility: could a competent stranger repeat this from what you wrote?
- Specify inclusion and exclusion criteria at the level of the individual participant
- State the sample size calculation with every input, not just the result

- Name the statistical tests against each objective, and say how missing data and confounding will be handled
- Pre-specify the analysis plan; deciding tests after seeing the data invites accusations of p-hacking
- For qualitative work, state the analytic approach, who will code, and how disagreements will be resolved
- Use the relevant reporting checklist while writing rather than afterwards: CONSORT for trials, STROBE for observational studies, COREQ or SRQR for qualitative work, PRISMA for reviews

IN PRACTICE

Reporting checklists are usually treated as a publication chore. Used at the proposal stage they function as a design checklist, and they surface the omissions reviewers would have found later.

Timeline, budget and feasibility

- Build the timeline backwards from the deadline, and add contingency — ethical approval alone commonly takes one to three months
- A Gantt chart makes dependencies visible: recruitment cannot begin before approval, analysis cannot begin before data cleaning
- Budget lines typically include personnel, transport, materials, participant compensation, data collection tools, software, transcription and translation, dissemination
- Transcription is routinely underestimated — an hour of interview takes roughly four to six hours to transcribe
- Justify every line; unexplained round numbers attract scrutiny
- State what you will do if recruitment is slower than planned — reviewers ask

Rigour and limitations

- Rigour is designed in, not asserted at the end
- Quantitative: adequate power, valid instruments, control of confounding, pre-specified analysis, appropriate handling of missing data
- Qualitative: audit trail, reflexivity, triangulation, thick description, transparent coding
- Limitations should name the threat, its likely direction, and what you did to reduce it
- "Time and financial constraints" is not a limitation; a convenience sample that may over-represent urban, literate respondents is
- Do not list limitations so severe they undermine the study — if one does, redesign rather than disclose and proceed

Data management and dissemination

- A data management plan states what data are collected, how stored, who has access, how backed up, how long retained and what is shared
- Follow the 3-2-1 rule: three copies, two media, one off-site
- Version your files by date and never overwrite raw data — all cleaning happens in a script or a documented copy
- FAIR principles — Findable, Accessible, Interoperable, Reusable — increasingly required by funders
- Dissemination should reach participants and their communities, not only journals: policy briefs, community feedback meetings, radio, professional bodies
- Feeding results back to the community that provided them is an ethical obligation, and it is what makes the next study possible
- Plan dissemination in the proposal, and budget for it

Worked examples

A limitations section that earns marks instead of losing them

A student writes: "Limitations included time and financial constraints, and a small sample size. More research is needed."

1. Time and money are constraints on the researcher, not limitations of the evidence — they tell the reader nothing.
2. "Small sample" is unquantified: small relative to what, and with what consequence?
3. Name the specific threat: the sample came from one urban clinic.
4. State the likely direction: urban patients are more literate and closer to care, so adherence is probably overestimated.
5. State the mitigation: findings are presented as applying to urban facility attenders, not to the district population.
6. Rewritten: "Recruitment from a single urban clinic likely over-represents literate, higher-income patients and probably overstates adherence; results should be read as applying to urban facility attenders."

The point: A good limitation names the threat, its direction, and what you did about it. Examiners reward the researcher who saw the problem first.

Building the timeline backwards from the deadline

A student has 12 months and plans: "Months 1-2 proposal, 3-8 data collection, 9-10 analysis, 11-12 writing." It is already unrealistic.

1. Ethical approval commonly takes 1-3 months and cannot start until the proposal is finished — so fieldwork cannot begin in month 3.
2. For 20 interviews at 1 hour each, transcription alone is 80-120 hours of work, which is weeks not days.
3. Analysis cannot begin until cleaning is done, and cleaning always takes longer than planned.
4. Supervisors need time to read: allow two weeks per full draft, and expect at least three drafts.
5. Working backwards from submission: writing 3 months, analysis 2, transcription 1, fieldwork 3, approval 2, proposal 1 — that is 12 with no contingency.
6. So either reduce scope now, or accept the plan will slip.

The point: Build the schedule backwards from the deadline and put the dependencies in a Gantt chart. Most overruns are visible at proposal stage to anyone who does the arithmetic.

Key terms

Gantt chart

A bar chart showing project activities against time, making dependencies and overlaps visible.

STROBE

The reporting checklist for observational epidemiological studies.

COREQ

The 32-item reporting checklist for qualitative interview and focus group research.

Data management plan

A document specifying how data will be collected, stored, protected, retained and shared.

FAIR principles

Findable, Accessible, Interoperable, Reusable — standards for research data stewardship.

p-hacking

Trying analyses until a significant result appears, then reporting only that one.

Policy brief

A short, non-technical summary of findings and recommendations aimed at decision makers.

Activity

Assemble a two-page proposal skeleton for your own study: title, aim, three objectives, design, sampling and sample size, analysis plan, and three limitations each stating direction and mitigation. Exchange with a partner and mark it against the relevant reporting checklist.

Further reading

- **EQUATOR Network — reporting guidelines** — Free index of every checklist; find yours before you write.
- **von Elm et al. (2007) — STROBE statement** — What an observational study must report.
- **Tong, Sainsbury & Craig (2007) — COREQ** — The qualitative reporting standard.
- **Wilkinson et al. (2016) — FAIR Guiding Principles** — Increasingly cited in funder data requirements.

Self-check questions

1. The best test of whether a methods section is adequate is:
 - a. It exceeds 1,000 words
 - b. A competent researcher could reproduce the study from it alone
 - c. It cites the most recent literature
 - d. It names the statistical software
2. Which is a properly stated limitation?
 - a. Time and financial constraints limited the study.
 - b. The convenience sample was drawn from one urban clinic and likely over-represents literate, higher-income patients, probably overstating treatment adherence.
 - c. More research is needed in this area.
 - d. The sample size was small.
3. Choosing statistical tests after inspecting the data and reporting only the significant ones is:
 - a. Exploratory analysis
 - b. p-hacking
 - c. Sensitivity analysis
 - d. Intention to treat
4. A proposal budgets three days to transcribe 20 interviews averaging one hour. This is:

- a. Generous
 - b. Unrealistic — transcription typically takes four to six hours per recorded hour
 - c. About right
 - d. Irrelevant to reviewers
5. The 3-2-1 backup rule means:
- a. Three researchers, two supervisors, one committee
 - b. Three copies, on two types of media, with one stored off-site
 - c. Three years of retention, two backups, one archive
 - d. Three drafts before submission
6. Feeding findings back to the community that provided the data is best understood as:
- a. Optional courtesy if funds remain
 - b. An ethical obligation that should be planned and budgeted from the start
 - c. A task for the funder
 - d. Necessary only for clinical trials
-

Answers

1. **b.** Reproducibility is the standard. Length, citation currency and software names are incidental if a reader still could not repeat what you did.
 2. **b.** A useful limitation names the specific threat, its likely direction of bias, and lets the reader judge its effect. The others are generic statements that give the reader nothing to work with.
 3. **b.** Selecting analyses based on which produced significance inflates the false positive rate. Pre-specifying the analysis plan in the proposal is the protection against it; genuine exploratory work must be labelled as such.
 4. **b.** Twenty hours of audio implies roughly 80-120 hours of transcription. Underestimating this is one of the most common feasibility errors in qualitative proposals, and reviewers look for it.
 5. **b.** Three copies on two different media with one off-site protects against device failure, theft and fire — each of which has ended postgraduate projects that had only a laptop copy.
 6. **b.** Participants contributed time and risk on the understanding that knowledge would result. Returning findings respects that, and it is what sustains community willingness to take part in future research.
-